

MARASANIGE SAMARTH MAHENDRA

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EDUCATION

Northeastern University <i>Master's, Computer Science</i> Relevant coursework: Programming Design Paradigm, Database Management Systems, Algorithms, Natural Language Processing, Machine Learning, Foundation of Software Engineering, Mobile App development, Computer Systems	January 2024 - December 2025
Dayananda Sagar College of Engineering <i>Bachelor's, Computer Science</i>	August 2018 - July 2022

SKILLS

- Languages:** Python, Java, C/C++, SQL, NoSQL, JavaScript, TypeScript, HTML/CSS
- Frameworks & Libraries:** Django restframework, Flask, React.js, Node.js
- Databases:** PostgreSQL, Elasticsearch, MongoDB, Redis
- Cloud & DevOp:** AWS, Terraform, Docker, Kubernetes, Prometheus, Datadog, Celery
- Systems:** TCP/IP, Concurrency, Locking, Profiling, Caching, B-tree, async I/O, multi-threaded design
- System Design & Concepts:** Microservices, REST APIs, System Design, scalable backend, Data Structures & Algorithms, distributed systems
- Testing:** Cypress, Jest, JUnit, PyTest, CodeQL
- Data structures and Algorithms:** Solved 400+ problems on LeetCode (Top 1.6 % of all LeetCode accounts (by global rank))

PROFESSIONAL EXPERIENCE

Jobstats.fyi <i>Founder & Developer</i> - Built a real-time analytics platform visualizing 15K+ submissions from more than 3,700 candidates across 1K+ companies. - Architected scalable Flask + MongoDB pipelines with Datadog RUM observability, reaching 1K+ sessions in 48 hours organically. - Impact: Empowered the CS Careers community to explore 15,000+ submissions, benchmark outcomes, and uncover hiring trends by providing detailed analytics and insights through the platform	Remote <i>October 2025 - Present</i>
Draup <i>Associate Software development Engineer</i> - Maintained and enhanced core platform modules across digital stack, outsourcing, customer, and university features, ensuring reliable operation and scalability. - Designed a dynamic query framework enabling real-time data aggregation, improving chatbot performance by 60% and cutting new entity development time by 80%. - Revamped filtering logic to support boolean modifiers operators and nested conditions, unlocking complex query composition for users. - Developed 100+ modular REST APIs using Python, Django REST Framework, and PostgreSQL, with Redis caching and Celery workers for async task execution. - Implemented subscription-based access control to manage app-level permissions and feature visibility. - Led migration from PostgreSQL to Elasticsearch, enabling real-time analytics with 5× faster queries and higher scalability. - Applied query optimization (partitioning, indexing, view creation) to achieve 400% faster execution and 50% lower operational cost. - Built Datadog + AWS CloudWatch dashboards to monitor system health, reducing downtime from 4% to 1% and accelerating issue resolution by 75%.	Bengaluru, KA, India <i>August 2022 - November 2023</i>
<i>Associate Software development Engineer Intern</i> - Built Datadog dashboards, integrated AWS CloudWatch alarms to monitor platform health, reducing issue resolution time by 30% - Implemented caching to improve efficiency of image requests, resulting in a 70% reduction in load times - Developed self-running Jenkins jobs for database cleanup, cutting manual effort and improving efficiency by 25%	<i>April 2022 - July 2022</i>

PROJECTS & OUTSIDE EXPERIENCE

ButterDB — High-Performance Buffered B-Tree Key-Value Store (C, Python) <i>Python, (C, Concurrency, WAL)</i> - Built a concurrent single-node TCP key-value database from scratch with in-memory hashmap baseline, then replaced with custom on-disk B-Tree engine and write buffering (Be-Tree). - Implemented fixed-page file IO, per-node locking, Write-Ahead Logging (WAL), and crash recovery for durability and concurrency. - Designed benchmarking and metrics collection to evaluate throughput, latency, and flush efficiency under mixed read/write workloads. - Achieved ~8× higher insert throughput vs naïve B-Tree via batched buffer propagation and IO batching; code structured for systems-level performance experiments.	Remote <i>October 2025 - Present</i>
Open Jobs Analytics Platform – Backend Infra + Monitoring <i>Tech stack: Puppeteer, Python, Redis, celery, Mongodb, Grafana, Prometheus, GPT-4o</i> - Built Jobs, inspired by Levels.fyi, aggregating 500+ job postings across companies. - Designed a producer-consumer architecture using Celery, integrated with Prometheus and Grafana, achieving 99.9% uptime and efficient task monitoring. - Scraped dynamic web pages with Playwright and Puppeteer, harvesting 1000+ data points daily. - Developed a Python reverse proxy with router port-forwarding, reducing latency by 40%. - Automated CSS selector extraction using LLMs, processing HTML content in under 2 seconds per page, reducing new company onboarding effort by 90%.	Boston, MA, USA <i>December 2024 - Present</i>

- Enhanced stealth capabilities by experimenting with random headers, user agents, referrer headers, and OS configurations, reducing bot detection by up to 90%.
- [Link to project](#)

StackOverflow

February 2025 - April 2025

Full-Stack Q&A Platform (StackOverflow Clone) | TypeScript, JavaScript, React.js, Node.js, MongoDB, Cypress, Jest, CodeQL, DevOps, Front-End & Back-End Web Development

- Developed a full-stack Q&A web application inspired by Stack Overflow using React (frontend) and Node.js/Express with TypeScript (backend).
- Architected backend controllers and models following the MVC pattern, leveraging design patterns such as Facade, Validator, Strategy, and Factory to ensure modularity, maintainability, and scalability.
- Built comprehensive end-to-end and integration test suites using Cypress and Jest, automating user flows (e.g., posting questions/answers, commenting, voting) and achieving high code coverage and reliability.
- Enhanced user experience with a responsive, accessible, and theme-enabled UI, utilizing modern CSS and React context for state management and theming.
- Integrated CI with GitHub Actions, Cypress, and CodeQL for PR-based linting and coverage checks.
- [Link to project](#)

Real-Time AI Voice Assistant & Intelligent Agent Platform | OpenAI GPT-4o, Twilio, Discord, FastAPI

Boston, MA, USA

Tech stack: OpenAI GPT-4O, Twilio, Celery, FastAPI, Discord, Websockets, Render (deployment), Redis

April 2025 - May 2025

- Engineered a real-time AI voice assistant using Twilio + GPT-4o's streaming API, enabling phone-based two-way conversations with <500ms latency, VAD, and dynamic speech truncation support.
- Architected an intelligent agent system integrating OpenAI GPT-4 + Google Gemini with modular tools, dynamic function calling, and profile-aware responses via MongoDB and Discord.
- Built a scalable async backend with FastAPI + WebSockets, deployed on Render with a Celery worker handling long-running tool calls and real-time audio coordination.
- Deployed live demo via public phone number (833) 970-3274 using Twilio, showcasing job-query answering, system prompts with resume context, and cross-platform communication.
- [Link to project](#)

Stock Market Simulation Application - Java MVC

Boston, MA, USA

Java, Junit, MVC

February 2024 - April 2024

- Developed a Java MVC system managing investment portfolios, integrating a stock API, test coverage, and buy/sell on specified dates.
- Integrated data visualization to plot portfolio performance over time, utilizing bar and line charts for investment growth trends, stock gains/losses analysis, and moving average calculation.

Myocardium Wall motion and wall thickness map, Patent Pending :- App-no : 202341086278 (India) (Co-inventor/author)

Bengaluru, KA, India

Research Assistant | Python, Dicom, MRI, numpy, Tkinter, image processing

November 2021 - September 2023

- Generated a visual map from cine-series MRI scans to investigate heart wall motion, thickness, thickness difference, fibrosis during systole and diastole phases using dicom and python
- Employed custom algorithms to determine myocardium wall thickness and calculated distances in indistinct areas, increasing measurement precision by 50%.
- Optimized time complexity using numpy and multiprocessing, reducing execution time by factor of 60